

Blocking Confounding with Experiments

Conditioning-Based vs. Design-Based Approaches

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POLI 110 Tutorial

Today's Agenda

1. Conditioning-Based vs. Design-Based Approaches
2. Running Shoe Performance Study:
 - ▶ Design 1: Multivariate Regression
 - ▶ Design 2: Difference-in-Differences
 - ▶ Design 3/4: Within-Runner Comparison
3. Comparative Analysis
4. Discussion

Conditioning-Based vs. Design-Based

Conditioning-Based	Design-Based
Statistically adjust for <i>observed</i> confounders	Leverage random or quasi-random variation
Examples: Regression, Matching, Propensity Scores	Examples: RCTs, DiD, IV, RDD
Limitation: Cannot handle unobserved confounders	Limitation: Requires strong assumptions
Vulnerable to omitted variable bias	May be infeasible or unethical

Conditioning-Based Methods

Core Idea: Control for observed confounders statistically

Example: Multiple Regression

$$Y_i = \alpha + \beta X_i + \gamma_1 U_{i1} + \gamma_2 U_{i2} + \cdots + \varepsilon_i$$

Key Assumption:

- ▶ All relevant confounders are *observed and included*
- ▶ Model is correctly specified

Problem: What about confounders we didn't measure?

Design-Based Methods

Core Idea: Structure data collection so treatment is "as if" random

Examples:

- ▶ **RCT:** Randomly assign treatment
- ▶ **Difference-in-Differences (DiD):** Compare before/after changes across groups

$$\widehat{ATE}_{DiD} = (\bar{Y}_{treat,after} - \bar{Y}_{treat,before}) - (\bar{Y}_{control,after} - \bar{Y}_{control,before})$$

- ▶ **Instrumental Variables (IV):** Use exogenous variation in X
- ▶ **Regression Discontinuity (RDD):** Exploit cutoff rules

Advantage: Can address unobserved confounders (if assumptions hold)

Do new running shoes improve marathon performance?

- ▶ $X_i = 1$ if runner uses new shoe, 0 otherwise
- ▶ $Y_i =$ finishing time (minutes)

Challenge: Runners who buy expensive new shoes might differ from those who don't

- ▶ More serious about running?
- ▶ Better training?
- ▶ Wealthier?

Design 1: Multivariate Regression

Method: Cross-sectional regression with controls

$$Y_i = \alpha + \beta X_i + \gamma_1 \text{Age}_i + \gamma_2 \text{Gender}_i + \gamma_3 \text{Training}_i + \varepsilon_i$$

Classification

Conditioning-Based

Confounders Addressed:

- ▶ Age, gender, training volume, race difficulty, weather

Confounders NOT Addressed:

- ▶ Unobserved: motivation, training technique changes, sponsorship

Key Assumption: No omitted variable bias

Design 2: Difference-in-Differences

Method: Track same runners in same races before/after shoe switch

$$\widehat{ATE}_{DiD} = (\overline{Y}_{\text{shoe,after}} - \overline{Y}_{\text{shoe,before}}) - (\overline{Y}_{\text{no shoe,after}} - \overline{Y}_{\text{no shoe,before}})$$

Classification

Design-Based

Confounders Addressed:

- ▶ Time-invariant runner traits (ability)
- ▶ Time-varying race features (difficulty, weather)

Confounders NOT Addressed:

- ▶ Time-varying factors correlated with shoe adoption (new training regimen)

Design 2: DiD Assumption

Parallel Trends Assumption:

Without the new shoe, both groups would have followed the same trend:

$$\mathbb{E}[Y(0)_{\text{after}} - Y(0)_{\text{before}} \mid \text{Shoe} = 1] = \mathbb{E}[Y(0)_{\text{after}} - Y(0)_{\text{before}} \mid \text{Shoe} = 0]$$

In plain English:

- ▶ If both groups had NOT switched to the new shoe...
- ▶ Their running times would have changed by the same amount

Design 3/4: Within-Runner Comparison

Method: Compare each runner before and after switching

$$\widehat{\text{Effect}}_i = Y_{i,\text{after}} - Y_{i,\text{before}}$$

Average across all switchers:

$$\widehat{\text{ATE}} = \frac{1}{n} \sum_{i=1}^n (Y_{i,\text{after}} - Y_{i,\text{before}})$$

Classification

Design-Based

Confounders Addressed: Time-invariant runner characteristics

Confounders NOT Addressed: Time-varying factors (new diet, training)

Comparing the Four Designs

Design	Type	Confounders Addressed	Key Assumption
Design 1: Regression	Conditioning	Observed: age, gender, training, weather	All confounders observed
Design 2: DiD	Design	Time-invariant runner traits; race features	Parallel trends
Design 3/4: Within-Runner	Design	Time-invariant runner traits	No time-varying confounders

Key Insight:

- ▶ Conditioning methods adjust for what we *observe*
- ▶ Design methods use structure to difference out confounders
- ▶ Design 2 is most powerful (controls both runner & race factors)

Key Takeaways

1. **Confounding** threatens causal inference when U affects both X and Y
2. **Randomization** breaks the link between confounders and treatment
3. **Two Approaches:**
 - ▶ **Conditioning-Based:** Control for observed confounders
 - ▶ **Design-Based:** Exploit random/quasi-random variation
4. Each approach addresses **different types** of confounders and requires **different assumptions**
5. The running shoe example shows how different designs handle the same research question

Questions for Discussion

1. Can you think of other confounders in the running shoe study?
2. Which design is most convincing? Why?
3. When might conditioning-based methods be the only option?
4. How would you test the parallel trends assumption in Design 2?